

CBSE Class 10 Mathematics — Probability

— Solutions

1. 1. If the probability of an event E occurring is 0.65, what is the probability of the event 'not E '?

Answer: (B)

Solution:

1. Recall the property of complementary events: $P(E) + P(\text{not } E) = 1$.
2. Substitute the given value $P(E) = 0.65$.
3. Solve for $P(\text{not } E) = 1 - 0.65$.

Derivation:

$$P(\bar{E}) = 1 - P(E) = 1 - 0.65 = 0.35$$

Probability — Complementary events

2. 2. The probability of winning a game is $\frac{7}{12}$. What is the probability of losing the game?

Answer: (C)

Solution:

1. The event 'losing' is the complement of 'winning'.
2. Use the formula $P(\text{losing}) = 1 - P(\text{winning})$.
3. Calculate $1 - \frac{7}{12}$.

Derivation:

$$P(\text{Losing}) = 1 - \frac{7}{12} = \frac{12 - 7}{12} = \frac{5}{12}$$

Probability — Complementary events

3. 3. A bag contains balls of different colours. If the probability of picking a red ball is $\frac{2}{9}$, what is the probability that the ball picked is NOT red?

Answer: (D)

Solution:

1. The event of picking a ball that is not red is the complement of picking a red ball.
2. Apply $P(\text{not red}) = 1 - P(\text{red})$.
3. Calculate $1 - \frac{2}{9}$.

Derivation:

$$P(\text{not Red}) = 1 - \frac{2}{9} = \frac{9 - 2}{9} = \frac{7}{9}$$

Probability — Complementary events

4. **4.** If $P(E) = 0.07$, find $P(\text{not } E)$.

Answer: (A)

Solution:

1. Use the complement rule $P(\text{not } E) = 1 - P(E)$.
2. Subtract 0.07 from 1.

Derivation:

$$P(\bar{E}) = 1 - 0.07 = 0.93$$

Probability — Complementary events

5. **5.** A student solves a problem correctly with a probability of 0.85. What is the probability that the student solves the problem incorrectly?

Answer: (C)

Solution:

1. The event of solving incorrectly is the complement of solving correctly.
2. Use $P(\text{incorrect}) = 1 - P(\text{correct})$.
3. Calculate $1 - 0.85$.

Derivation:

$$P(\text{incorrect}) = 1 - 0.85 = 0.15$$

Probability — Complementary events

6. **6.** Assertion (A): If the probability of winning a game is 0.7, then the probability of losing the game is 0.3. Reason (R): The sum of the probabilities of all complementary events is 1.

Answer: (A)

Solution:

1. Check if the events winning and losing are complementary.
2. Check if the sum of probabilities equals 1.
3. Verify the calculation $1 - 0.7 = 0.3$.

Derivation:

$$\begin{aligned} P(E) + P(\text{not } E) &= 1 \\ \implies 0.7 + 0.3 &= 1.0 \end{aligned}$$

Probability — complementary events

7. **7.** Assertion (A): The probability of an event happening is 0.45, then the probability of the event not happening is 0.65. Reason (R): For any event E, $P(\bar{E}) = 1 - P(E)$.

Answer: (D)

Solution:

1. Identify the formula $P(\bar{E}) = 1 - P(E)$.
2. Calculate $1 - 0.45 = 0.55$.

3. Since $0.55 \neq 0.65$, Assertion is false.

Derivation:

$$P(\bar{E}) = 1 - 0.45 = 0.55$$

Probability — complementary events

8. **8.** Assertion (A): If a fair coin is tossed, the probability of getting a head is 0.5 and the probability of not getting a head is 0.5. Reason (R): Getting a head and not getting a head are complementary events.

Answer: (A)

Solution:

1. Identify the outcomes of a coin toss as Head and Tail.
2. Understand that not getting a head is equivalent to getting a tail.
3. Verify $P(H) = 0.5$ and $P(\bar{H}) = 1 - 0.5 = 0.5$.

Derivation:

$$P(H) = 1/2 = 0.5; P(\text{not } H) = 1 - 0.5 = 0.5$$

Probability — complementary events

9. **9.** Assertion (A): If the probability of selecting a red ball from a bag is $2/7$, the probability of not selecting a red ball is $5/7$. Reason (R): The probability of an impossible event is 1.

Answer: (C)

Solution:

1. Check the complementary probability: $1 - 2/7 = 5/7$. Assertion is true.
2. Identify that the probability of an impossible event is 0, not 1. Reason is false.

Derivation:

$$P(\bar{E}) = 1 - 2/7 = 5/7$$

Probability — complementary events

10. **10.** Assertion (A): If $P(E) = 0.12$, then $P(\bar{E}) = 0.88$. Reason (R): $P(E) + P(\bar{E}) = 1$.

Answer: (A)

Solution:

1. Apply the formula $P(\bar{E}) = 1 - P(E)$.
2. Calculate $1 - 0.12 = 0.88$.
3. Confirm that the reason provided is the definition of complementary events.

Derivation:

$$0.12 + 0.88 = 1.00$$

Probability — complementary events

11. 11. A box contains 90 discs which are numbered from 1 to 90. If one disc is drawn at random from the box, find the probability that it bears (i) a two-digit number, (ii) a perfect square number, (iii) a number divisible by 5. Use the concept of complementary events to verify the probability of the event 'not getting a perfect square'.

Answer:

$$(i) 9/10, (ii) 1/10, (iii) 1/5; \text{Complement} : 9/10$$

Solution:

1. Identify the total number of outcomes as 90.
2. List perfect squares between 1 and 90: {1, 4, 9, 16, 25, 36, 49, 64, 81}. Total = 9.
3. Calculate $P(\text{Perfect Square}) = 9/90 = 1/10$.
4. Use the complementary event formula: $P(\text{Not } E) = 1 - P(E)$.
5. Calculate $P(\text{Not Perfect Square}) = 1 - 1/10 = 9/10$.

Derivation:

$$\begin{aligned} P(E) &= \frac{9}{90} = \frac{1}{10} \\ P(\bar{E}) &= 1 - P(E) \\ P(\bar{E}) &= 1 - \frac{1}{10} = \frac{9}{10} \end{aligned}$$

Probability — Complementary events

12. 12. A game consists of tossing a one-rupee coin 3 times and noting its outcome each time. Hanif wins if all the tosses give the same result i.e., three heads or three tails, and loses otherwise. Calculate the probability that Hanif will lose the game using the complementary event property.

Answer:

$$3/4$$

Solution:

1. List all possible outcomes: {HHH, HHT, HTH, THH, HTT, THT, TTH, TTT}. Total = 8.
2. Identify favorable outcomes for winning: {HHH, TTT}. Count = 2.
3. Calculate $P(\text{Win}) = 2/8 = 1/4$.
4. Identify the event of losing as the complement of winning.
5. Apply $P(\text{Lose}) = 1 - P(\text{Win})$.

Derivation:

$$\begin{aligned} P(\text{Win}) &= \frac{2}{8} = \frac{1}{4} \\ P(\text{Lose}) &= 1 - P(\text{Win}) \\ P(\text{Lose}) &= 1 - \frac{1}{4} = \frac{3}{4} \end{aligned}$$

Probability — Complementary events

- 13. 13.** A bag contains 5 red balls and some blue balls. If the probability of drawing a blue ball is double that of a red ball, find the number of blue balls in the bag. Further, if a ball is drawn at random, what is the probability of not drawing a red ball?

Answer:

$$10 \text{ blue balls; } P(\text{Probability of not red}) = 2/3$$

Solution:

1. Let the number of blue balls be x . Total balls = $5 + x$.
2. $P(\text{Red}) = 5/(5 + x)$, $P(\text{Blue}) = x/(5 + x)$.
3. Given $P(\text{Blue}) = 2 * P(\text{Red})$, solve for x : $x/(5+x) = 2 * (5/(5+x))$.
4. Solving results in $x = 10$.
5. Calculate $P(\text{Not Red}) = 1 - P(\text{Red})$.

Derivation:

$$\begin{aligned} \frac{x}{5+x} &= 2 \times \frac{5}{5+x} \\ \implies x &= 10 \\ P(\text{Red}) &= \frac{5}{15} = \frac{1}{3} \\ P(\text{Not Red}) &= 1 - \frac{1}{3} = \frac{2}{3} \\ &\text{Probability — Complementary events} \end{aligned}$$

- 14. 14.** Two dice are thrown simultaneously. Find the probability of getting (i) a sum of 8, (ii) a doublet, (iii) a sum greater than 9. Use the complementary event concept to find the probability of getting a sum not equal to 8.

Answer:

$$(i) 5/36, (ii) 6/36, (iii) 6/36; \text{ Complement : } 31/36$$

Solution:

1. Total outcomes for two dice = $6 * 6 = 36$.
2. Identify outcomes for sum 8: $\{(2,6), (3,5), (4,4), (5,3), (6,2)\}$. Count = 5.
3. $P(\text{Sum 8}) = 5/36$.
4. Use complement property for 'Sum not 8': $P(\text{Not 8}) = 1 - P(8)$.
5. Final calculation: $1 - 5/36 = 31/36$.

Derivation:

$$\begin{aligned} P(\text{Sum} = 8) &= \frac{5}{36} \\ P(\text{Sum} \neq 8) &= 1 - P(\text{Sum} = 8) \\ P(\text{Sum} \neq 8) &= 1 - \frac{5}{36} = \frac{31}{36} \\ &\text{Probability — Complementary events} \end{aligned}$$